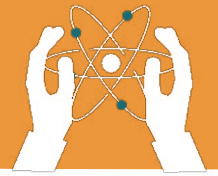


UNIT 7

INTRODUCTION TO PHYSICS

Learning objectives:

- Understand what physics is and its significance in explaining natural phenomena.
- Explore branches of physics, such as mechanics, thermodynamics, and electricity.
- Learn the scientific method and its role in studying and solving real-world problems.

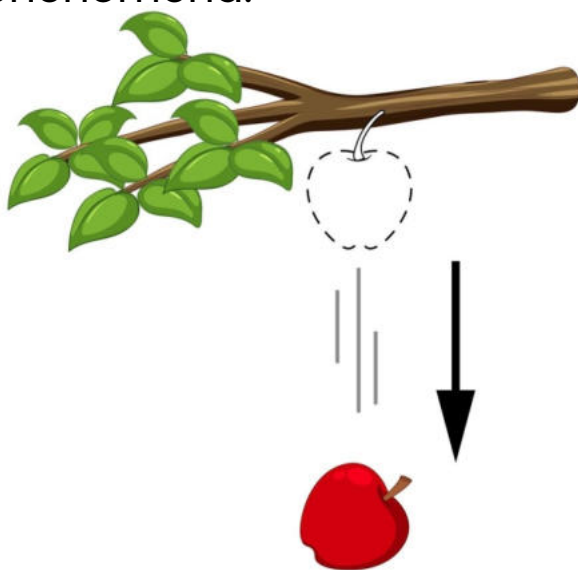


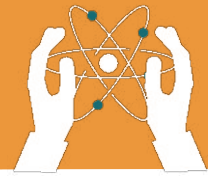
7.1 What Is Physics?

Physics is the study of the natural world around us. It explains how and why things move, how energy works, and how forces interact.

Physics is one of the oldest and most fundamental sciences, helping us understand the behaviour of matter, energy, space, and time. For example, physics helps us understand: why a ball falls to the ground? How electricity powers devices? Or why the stars shine?

Physics is often called the "**mother of all sciences**" because it provides the foundation for other branches of science like chemistry and biology. It uses observations, experiments, and mathematical calculations to explain natural phenomena.





7.2 Branches of Physics

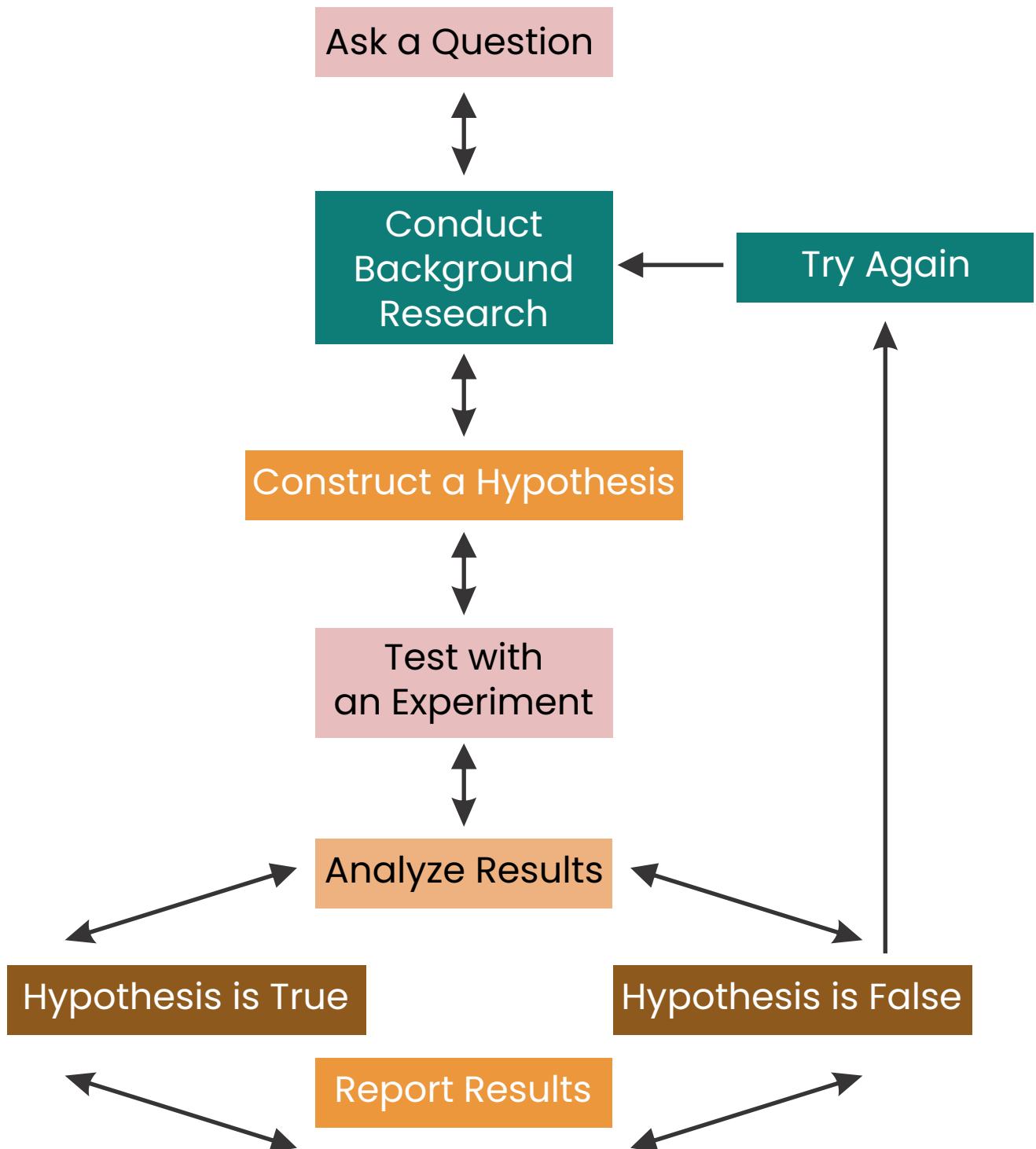
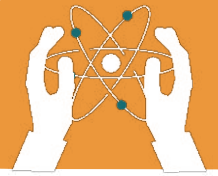
Physics has several branches that focus on different aspects of the physical world:

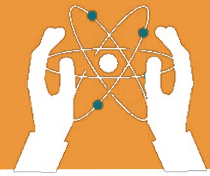
- 1. Mechanics:** This studies motion and forces, like how cars move or how planets orbit the sun.
- 2. Thermodynamics:** It deals with heat and energy, such as how refrigerators work.
- 3. Electricity:** It explores electrical charges, circuits, and devices like fans or mobile phones.
- 4. Magnetism:** This studies magnetic fields, like how compasses work.
- 5. Optics:** It explores light and vision, such as lenses in glasses or microscopes.
- 6. Sound:** This studies sound waves and their properties, like music or echoes.

These branches help us understand and explain everyday phenomena.

7.3 The Scientific Method in Physics

The scientific method is the process physicists use to study the natural world. It includes the following steps:





1. Observation: Carefully watching and noticing events in nature is termed as observation

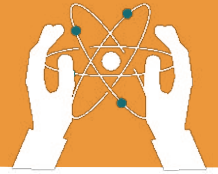
2. Hypothesis: Making a guess or prediction about why something happens is known as hypothesis.

3. Experimentation: Testing the hypothesis through experiments termed as experimentation.

4. Analysis: Reviewing the results of the experiment is called as Analysis

5. Conclusion: Deciding whether the hypothesis was correct or not is called as conclusion.

For example, if you want to know why a ball falls, you observe it, predict that gravity pulls it down, test it by dropping the ball, and conclude your prediction is correct.



7.4 Importance of Physics in Science, Technology, and Society

Physics plays a crucial role in shaping the world we live in. It has led to the development of modern technologies like mobile phones, electricity, airplanes, and medical equipment. Physics also helps us explore space, understand the universe, and predict natural phenomena like earthquakes.

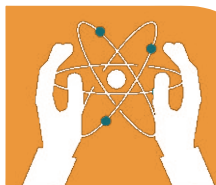
Physics improves our quality of life by creating innovative solutions for transportation, communication, and energy production.

For example:

Role of Physics in Everyday Life

- **Household Applications like** electricity, appliances, and sound systems.
- **Transportation uses** principles of motion and mechanics in vehicles and aircraft.
- **Healthcare uses** Medical imaging (X-rays, MRI) and radiation therapy.





EXERCISES

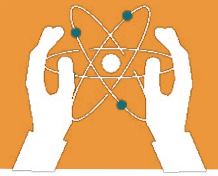
Fill in the blanks.

1. _____ is the study of motion, forces, and energy.
2. The scientific method starts with _____.
3. _____ is the branch of physics that deals with heat and energy.
4. Physics is called the _____ of all sciences.
5. The process of testing a hypothesis is called an _____.
6. _____ helps us understand the motion of planets.
7. Optics is the branch of physics that studies _____.
8. Physics has improved our lives by advancing _____ and communication.

Tick(✓) for True and cross (x) for False statements

1. Physics is the study of living organisms.
2. Thermodynamics deals with the motion of objects.
3. The scientific method includes observing and experimenting.
4. Optics is the study of light and its properties.
5. Physics has no role in modern technology.
6. The conclusion is the final step in the scientific method.
7. Magnetism is the study of heat and energy.
8. Mechanics deals with forces and motion.

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Choose the correct answer

1. Physics is the study of:

- a) Living organisms
- b) The natural world
- c) Planets only
- d) Stars only

2. The branch of physics that deals with electricity is:

- a) Optics
- b) Mechanics
- c) Magnetism
- d) Electricity

3. The first step in the scientific method is:

- a) Experimentation
- b) Hypothesis
- c) Observation
- d) Conclusion

4. Which branch of physics studies sound?

- a) Mechanics
- b) Optics
- c) Sound
- d) Thermodynamics

5. Thermodynamics is the study of:

- a) Electricity
- b) Heat and energy
- c) Forces
- d) Light

6. The process of forming a prediction in the scientific method is called:

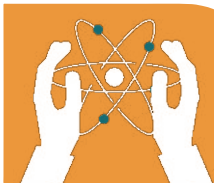
- a) Observation
- b) Experimentation
- c) Hypothesis
- d) Analysis

7. Physics has helped develop:

- a) Smartphones
- b) Cars
- c) Rockets
- d) All of the above

8. Magnetism is the study of:

- a) Light
- b) Motion
- c) Magnetic properties
- d) Sound



Answer the Following Questions:

Short-Answer Questions

- 1. What is physics?**
- 2. Name three branches of physics.**
- 3. What is scientific method?**

Long-Answer Questions

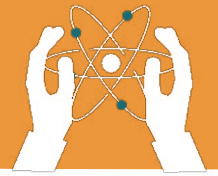
- 1. What are the main branches of physics?**

Reasoning Questions Answer

- 1. Why is physics called the mother of all sciences?**

Short Note

- 1. Write a short note on importance of physics in daily life.**



ACTIVITY

- **Explore Gravity:** Drop two different objects, like a pen and a paper, from the same height. Record your observations about their motion.



Critical Thinking

Why do you think physics is essential for advancements in medicine and transportation? Provide examples to support your answer.